SMART Weight Check

Weigh at source. Check every tonne.

An independent scale in the Europress compactor checks every lift and every per-tonne invoice line, and a human approves every proposal.

Since AI Hackathon 2026 · Europress challenge · all data synthetic, not Europress data

Customers pay per tonne they never see



CMP-00046 · cardboard · Lohja · Jul–Sep 2026. Top: fill % · bottom: container weight, kg (simulated). Dashed lines: trips without a lift.

July: the fill reading stuck at 88 % while the scale read ~0 kg. Four trucks came for nothing.
Nobody did anything wrong. Nobody could see it.

2.3×

€ per tonne vs the contract benchmark: €258 vs €110 (estimated)

€346

for 4 wasted trips while the fill reading was stuck (observed)

6–25 %

fill at three customer-ordered lifts in September (simulated)

Weigh, check, propose: a human decides



1 · Weigh

Four load cells under the container. Gross before the lift minus tare after = collected kg. A free zero check on every lift.



2 · Check

Every lift and every per-tonne invoice line against device weight, the operator's ticket and the contract. Combined loads and side waste are explained, not blamed.



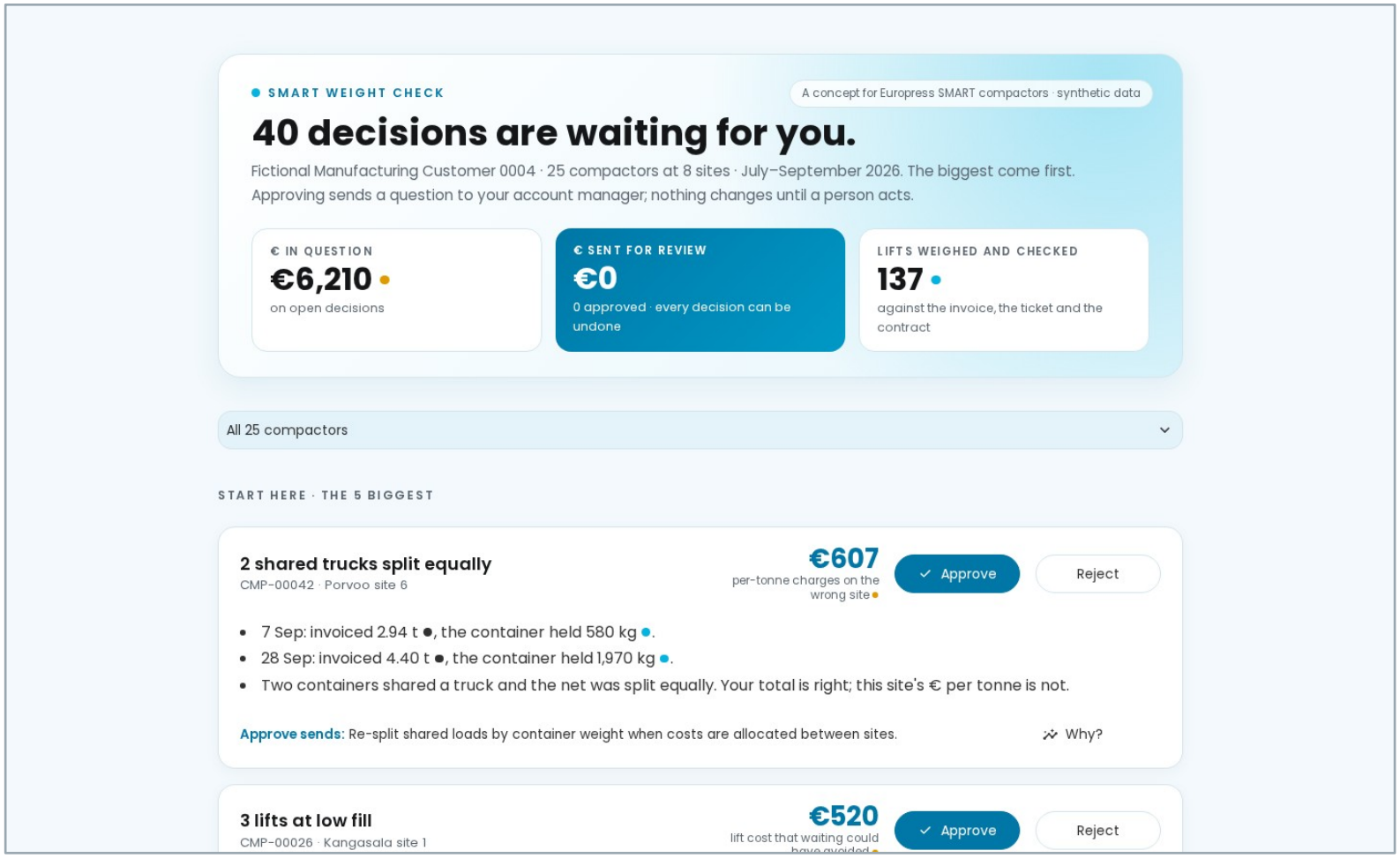
3 · Propose

Each difference becomes a proposal with a reason and a check before acting. Approve, reject, undo. Nothing executes by itself.

Every number carries its evidence:

observed read from a record	**simulated** SMART device data	**estimated** our calculation	**hypothetical** an assumption

Live demo: every lift checked

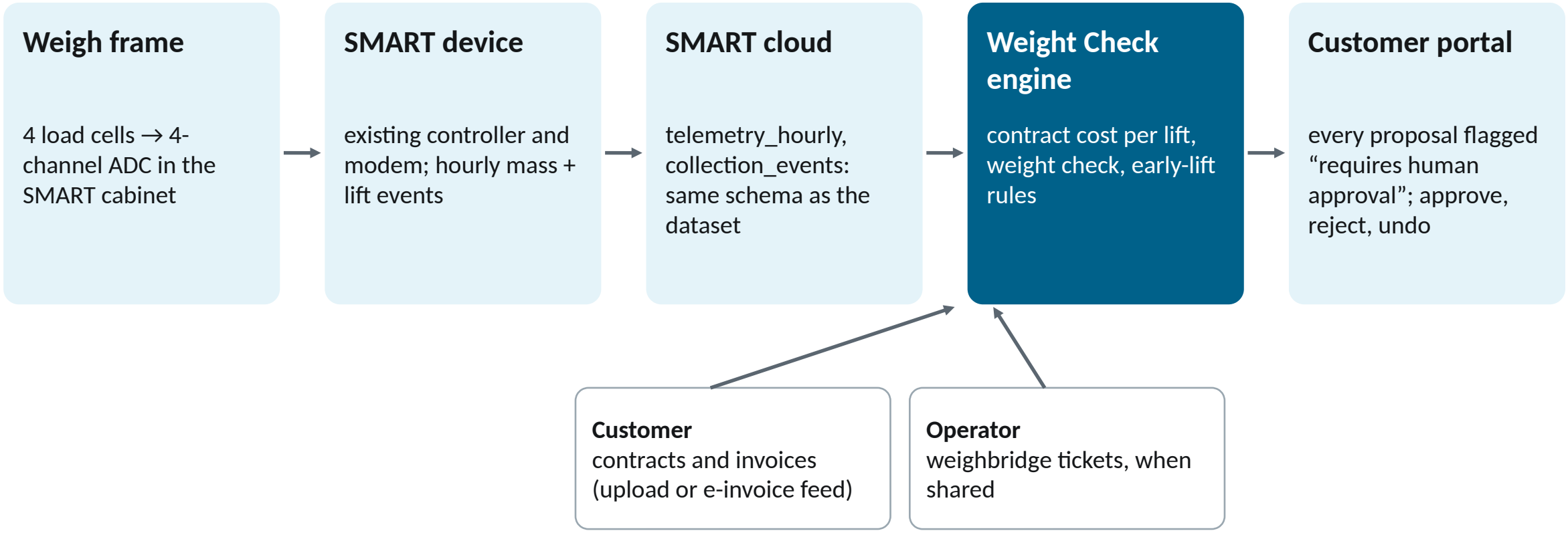


Demo path

1. CMP-00046: the stuck fill line vs the flat weight line; 2.3× € per tonne
2. CMP-00042: invoiced 2.94 t, container held 580 kg. A shared truck, split equally: re-split by device weight
3. CMP-00047: no ticket, invoice 425 kg above the scale: ask for the ticket
4. Approve, then undo

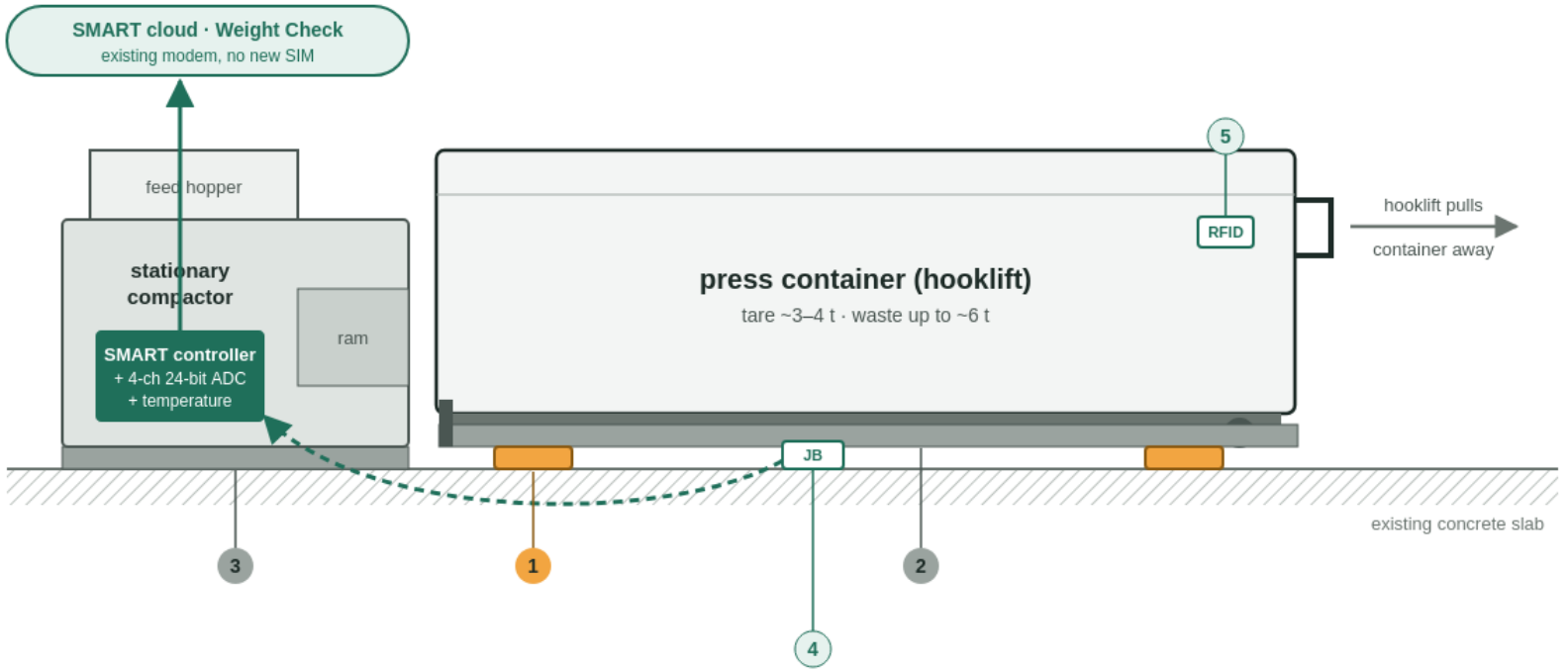
341 lifts checked: 205 agree · 38 explained · 4 billing questions · 13 devices: check the scale

From load cell to customer decision



Runs on what SMART already has: the cabinet, the modem, the telemetry pipeline. The weight is the customer's own measurement; it is shared with the operator only with consent.

A weigh frame under the container



How one lift is weighed: stable gross reading with the ram idle, minus the tare when the empty container comes back. While the container is away the frame must read zero: a free zero check on every lift. Four separate cells reveal ice or debris under one corner.

σ 1.5 %

design target; ~1.0 % accuracy budget (estimate)

€1,200

factory option · €3,400 installed retrofit (hypothetical BOM)

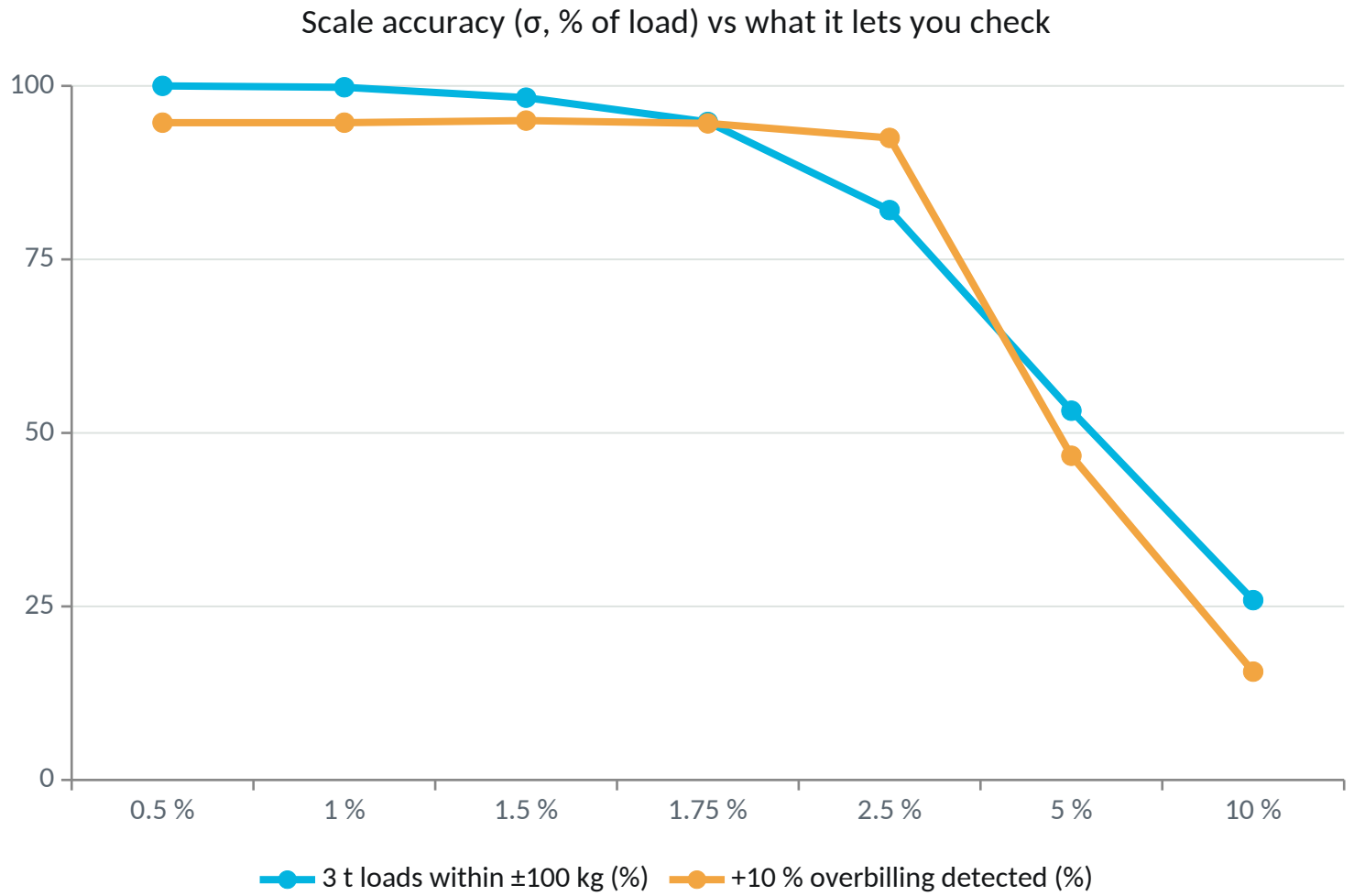
Self-checking

auto-zero every time the container leaves; span checked against the operator's tickets

Validation

2-week yard test with water tanks → 20-site pilot, 3 months → winter

Accuracy decides what weight is worth



Competition data, 10,342 shared single-load tickets; examples/weighing_accuracy_value.py (estimated)

≤ 1.75 %

keeps ≥ 95 % of 3 t loads within ±100 kg: the challenge target

› 2.5 %

and checking the operator's tonnage stops working

7 %

best estimate without a scale (ML or rotating kit): a free value scan, not an invoice check

€15 per device-month, priced from value

€36

median realisable value per device-month (estimated): the customer keeps 2.4× the fee

2.0 years

payback of onboarding at €15 and 25 % adoption (hypothetical)

Yearly profit after onboarding, € (hypothetical, 52 synthetic customers)

price / adoption	10 %	25 %	40 %
€10	943	2,358	3,773
€15	2,148	**5,370**	8,592
€25	1,735	4,338	6,941

Price matters most. €15 maximises fee revenue; at €10 small customers cost more to support than they pay (6.5-year payback).

Sell to fleets of 6+ devices

16+ devices: ~€4.0 k profit per customer-year; 2–5 devices lose money at €50 support

Weigh where it pays

factory option pays back in 2.9 years at the median device, retrofit in 8.2

Hypothesis: per-tonne full service

3 % margin ≈ €156 k/year on this sample; needs verified weight and a new role

What we have not proven, and next steps

Not proven

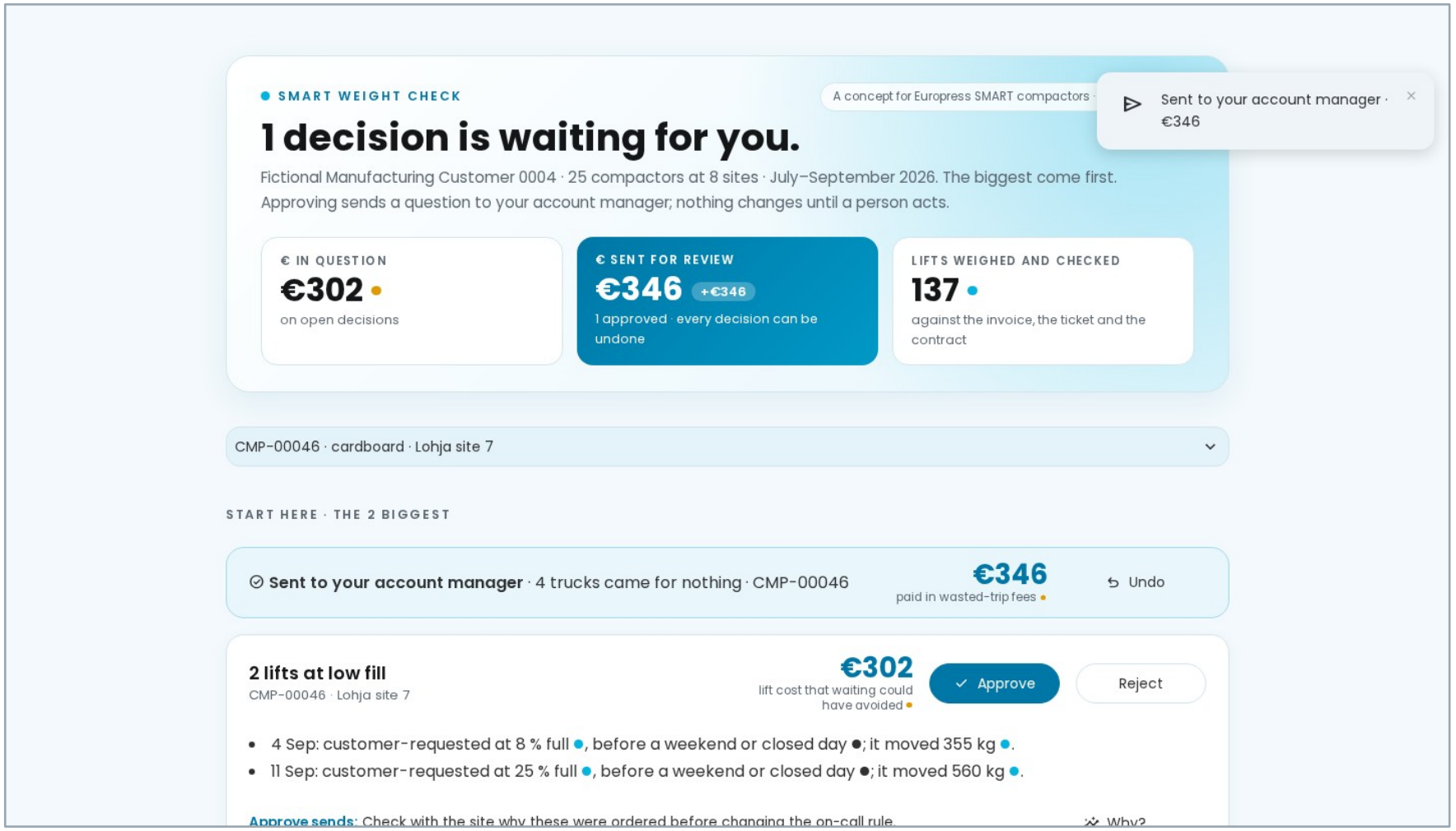
- that customers pay €15 per device-month: no interviews yet
- that the savings are real: synthetic world, hypothetical tariffs
- that the frame reaches σ 1.5 % on a real compactor in winter
- the hardware cost: indicative BOM, no supplier quotes
- that operators accept device weight as evidence

Next steps

1. 10 customer interviews: willingness to pay
2. 2-week yard test: water-tank steps, hooklift cycles
3. 20-site pilot, 3 months, vs weighbridge tickets

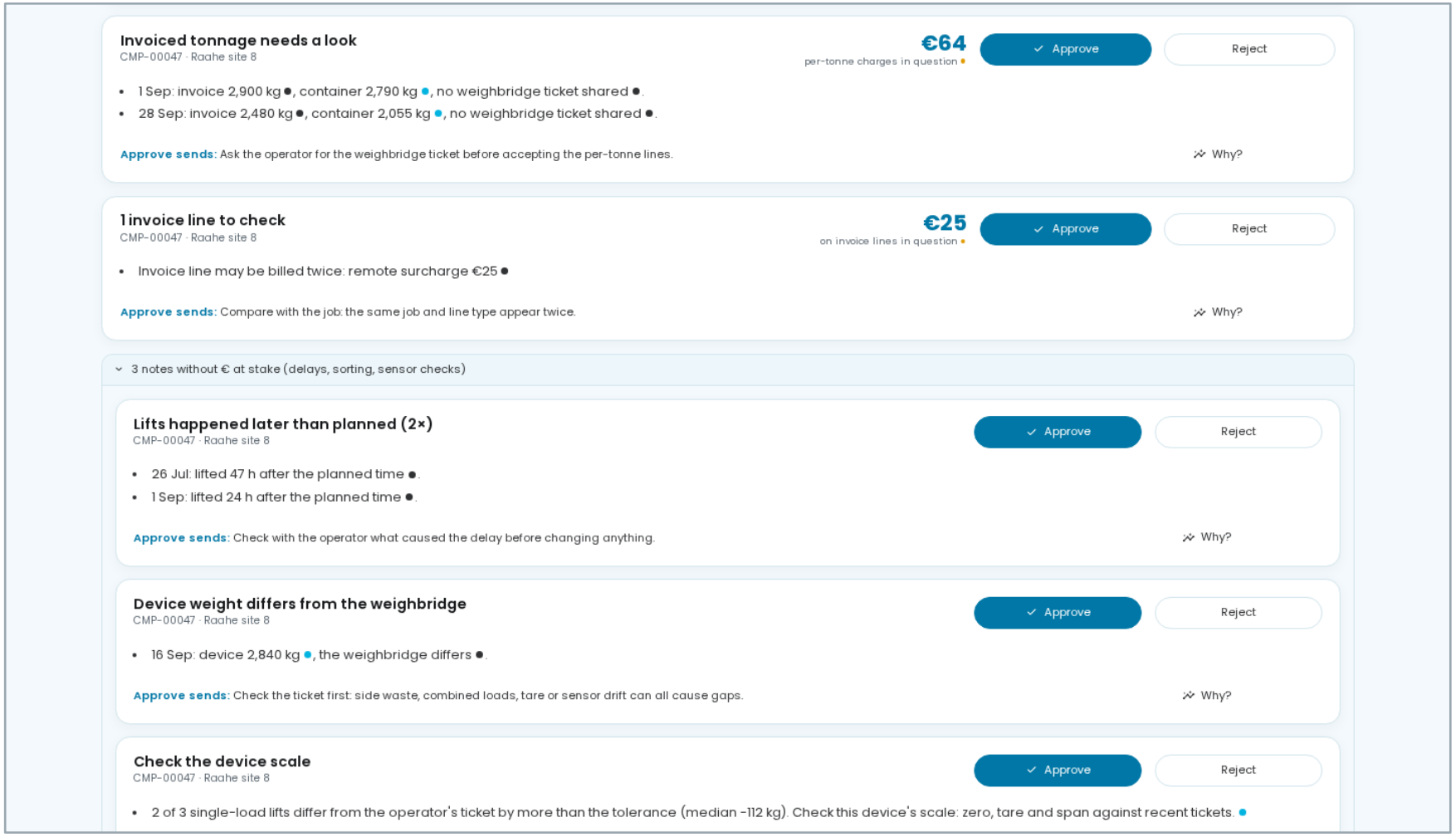
Weigh at source. Check every tonne.

Backup: proposals, a human decides



CMP-00046: proposals with a reason and a check; one approved, with Undo. Weight rules add the early lifts on 4 and 11 September.

Backup: a lift without a ticket



CMP-00047: two lifts invoiced without a shared ticket differ from the device weight, so the proposal is to ask for the ticket. The scale-check note points at the device, not the operator.

Run it, sources and licences

Run the demo

```
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev,app]"
python baselines/b1_economic_calculator.py data/quickstart/public my_task_a.csv
python baselines/b2_rules_anomaly.py data/quickstart/public my_task_b.csv
streamlit run app/app.py
```

Where everything is

- submission/1_WEIGHING_CONCEPT.md: sensor, BOM, calibration, validation
- submission/2_SOFTWARE_DEMO.md: walkthrough, SMART integration
- submission/3_BUSINESS_CASE.md: value, pricing, sensitivity, limits
- submission/THIRD_PARTY.md: every library, dataset and service, with licence
- notes/FINDINGS.md: every number with its source table or script

Data: Europress × Since AI synthetic dataset v1.0.1 (use per the hackathon rules). Code: Python with numpy, pandas, DuckDB, Streamlit, Altair, scikit-learn, matplotlib (BSD, MIT, Apache-2.0 and PSF-style licences, all allow commercial use). AI assistance: Claude (Anthropic) via Claude Code.